

THEJAS A.NAIR

PERSONAL DATA

Current address	Karingalayath house, Edoor Kannur, Kerala, 670 704
Nationality	Indian
Date and Place of birth	27-05-2000, Kerala, India.
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EDUCATION

Central University of Kerala, Kasargode MSc in Physics	<i>November 2021 - August 2023</i> CGPA: 7.75/10
Kannur University, Kannur BSc in Physics Mahatma Gandhi College	<i>June 2018 - April 2023</i> CGPA: 8.1/10
St Mary's Higher Secondary School Edoor, Kannur Plus Two	<i>July 2016 - May 2018</i> Overall Percentage: 97.17
St Mary's Higher Secondary School Edoor, Kannur SSLC	<i>April 2016</i> Secured A+ in all subjects

RESEARCH EXPERIENCE

Numerical Relativity using Einstein Toolkit (MSc Project) Under the supervision of Dr. Krishna Mohan Parattu, IIT Mandi Studied ADM formulation and aspects of Numerical Relativity. Studied the time evolution of the density of a neutron star using Einstein toolkit and compared the error propagation for various grid sizes.	<i>April 2023 - August 2023</i>
Galaxy Bars (MSc mini project) Under the supervision of Dr. Vinu Vikram, Central University of Kerala Studied the formation of galaxy bars and how bulges effect on bar formation.	
Determination of the wavelength of LEDs using the Newton ring experiment (BSc project) Under the supervision of C.V. Sandhya, Mahatma Gandhi College Calculated the wavelength of blue, green, and red LEDs using Newton's ring experiment	

COURSES AND WORKSHOPS

- **Summer school on gravitational wave astronomy** ICTS Bangalore *24 July 2023 - 4 August 2023*

- **International Conference on Functional Materials for Advanced Technology (ICFMAT-I)** Department of Physics, Central University of Kerala *2 January 2023 - 4 January 2023*
- Volunteered for the state level workshop on B.Sc experiments conducted by Department of physics Mahatma Gandhi College *December 2018*
- Recent trends in Raman Spectroscopy, University of Kerala *15 July 2020*
- 100 years of gravitational waves by Prof. Rai Weiss
World Science U
- Introduction to Classical Mechanics
National program on technology enhanced learning - IIT Hyderabad *July 2023 - October 2023*
- Group theory methods in Physics
National program on technology enhanced learning - IIT Bombay *January 2024 - April 2024*

RESEARCH INTEREST

My research interest is primarily in Gravitation and various related aspects; other interests include Quantum field theory and Cosmology.

ACADEMIC ACHIEVEMENTS

Qualified JEST and GATE ¹ with all India rank 496 and 1806 respectively	<i>2024</i>
Qualified IIT JAM ² with all India rank 1957	<i>2021</i>
Qualified CUCET ³ Central University of Kerala, Rank 4	<i>2021</i>
Central University of Karnataka, Rank 4	
Central University of Punjab, Rank 3	
Central University of Rajasthan, Rank 4	
Cochin University of Science and Technology CAT ⁴ , Rank 26	<i>2021</i>
Pondichery Central University entrance exam, Rank 47	<i>2021</i>
Qualified talent search examination by the Association of Physics Teachers Kerala (APT) with State-level Rank 41	<i>2021</i>
Selected as one of the 20 students by KTPI ⁵ for ART(Active research training) program to do MSc project under an external guide ⁶	<i>2022</i>

¹Two exams whose scores are considered for PhD admissions across premium institutes in India

²Joint admission test for Masters

³Central university common entrance tests

⁴Common admission test for MSc

⁵Kerala Theoretical Physics Initiative

⁶Later Cancelled because I wanted to work in a different topic

SKILLS

Fortran

Solved one-dimensional harmonic oscillator using the shooting method

Solved standard differential equations such as parabolic, hyperbolic, and elliptical using finite difference method.

Studied initial value problems and numerical methods such as tridiagonal, naive Gaussian elimination, and Monte Carlo.

Python

Solved differential equations using Runge-Kutta and Newton Raphson methods.

We used matplotlib and Pyplot to sketch the nature of simple graphs.

Origin software

I have worked on origin software during my 3-semester lab work (MSc). This included analyzing and calculating the parameters required from the graph to studying the XRD peak of the compound.

Languages

Malayalam, English, Hindi

ACTIVITIES

National Service Scheme (NSS) Mahatma Gandhi College

Participated in the 10-day NSS camp December 2018

Participated in the 3-day Nature camp 2019

Participated in the state-level model parliament competition (4th place)

Active member of flood relief programs in 2018

Quiz aficionado

Founder member of Quiz Club, Mahatma Gandhi College.

Second prize in inter-college quiz competition conducted by Department of Physics; Govt Brennen College Thalassery.

Outreach Program

Conducted a talk on the Apollo mission as part of its 50th anniversary (Mahatma Gandhi College 2019).

Participated and volunteered in a social awareness program conducted by KSSP⁷ as part of the annular solar eclipse (26 December 2019).

Sports

Participated in the inter-department Cricket and Football competitions.

References

Dr. Krishna Mohan Parattu

Assistant Professor

School of Physical sciences

IIT Mandi

Mandi, Himachal Pradesh

Email: krishna@iitmandi.ac.in

Prof. Vincent Mathew

Dean of Physical Sciences

Central University of Kerala

⁷Kerala Sasthra Sahithya Parishad

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Email: vincent@cukerala.ac.in

Dr. Vinu Vikram

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